

MODULE :- Selenium WebDriver

EXPERIMENT TITLE :- The Multi-Locator Challenge (Assignment 1)

1.Problem Statement :-

To automate the login process of a web application using Selenium WebDriver, identify the username field using an ID locator, the password field using a NAME locator, and the login button using an XPath locator. After login, verify that the browser is redirected to the expected inventory page.

2.Tools and Software Required :-

S. No.	Tool / Software	Purpose
1	Python	Programming language used to write the automation script
2	VS Code	IDE used to write and execute Python code
3	Google Chrome	Web browser used for automation
4	Selenium WebDriver	Automates the browser and interacts with web elements
5	Command Prompt / Terminal	Used to run the Python automation script
6	SauceDemo Website	Website used as the application under test

3.Concepts Used :-

Selenium WebDriver is used to automate and interact with the web browser.

ID Locator is used to identify the username field.

NAME Locator is used to identify the password field.

XPath Locator is used to identify the login button.

Assertions are used to verify that the login redirects to the expected inventory URL.

4.Source Code :-

File 1: assignment_1.py

```
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()

try:
    driver.get("https://www.saucedemo.com/")

    # Username using By.ID
    username = driver.find_element(By.ID, "user-name")
    username.send_keys("standard_user")
```

```
# Password using By.NAME
password = driver.find_element(By.NAME, "password")
password.send_keys("secret_sauce")

# Login button using By.XPATH
login_button = driver.find_element(
    By.XPATH,
    "//input[@id='login-button']"
)
login_button.click()

# Verify URL
assert "/inventory.html" in driver.current_url

print("PASS: Login successful")
print("PASS: URL contains /inventory.html")

finally:
    driver.quit()
```

5.Implementation Steps :-

- 1.Install Python and Selenium WebDriver.
- 2.Open the SauceDemo login page using Selenium.
- 3.Locate the username field using the ID locator.
- 4.Enter the valid username.
- 5.Locate the password field using the NAME locator.
- 6.Enter the valid password.
- 7.Locate the login button using an XPath locator.
- 8.Click the login button.
- 9.Verify that the current URL contains '/inventory.html'.
- 10.Close the browser after completing the test.

6.Output :-

The image displays two screenshots from a web browser window, with a terminal window overlaid at the bottom.

Top Screenshot: Swag Labs Login Page

The browser address bar shows `saucedemo.com`. The page title is "Swag Labs". The login form contains the following fields and buttons:

- Username field: `standard_user`
- Password field: (empty)
- Login button: (green)

Below the login form, a dark box lists the accepted usernames and the password for all users:

Accepted usernames are:	Password for all users:
<code>standard_user</code>	<code>secret_sauce</code>
<code>locked_out_user</code>	
<code>problem_user</code>	
<code>performance_glitch_user</code>	
<code>error_user</code>	
<code>visual_user</code>	

Bottom Screenshot: Swag Labs Inventory Page

The browser address bar shows `saucedemo.com/inventory.html`. The page title is "Swag Labs". The page displays a list of products:

- Sauce Labs Backpack**: `carry.allTheThings()` with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection. Price: `$29.99`. Button: `Add to cart`.
- Sauce Labs Bike Light**: A red light isn't the desired state in testing but it sure helps when riding your bike at night. Water-resistant with 3 lighting modes, 1 AAA battery included. Price: `$9.99`. Button: `Add to cart`.

Terminal Window

The terminal window shows the execution of a Python script:

```
PS C:\Users\catch\OneDrive\Desktop\wipro_assignments> python -u "c:\Users\catch\OneDrive\Desktop\wipro_assignments\assignment_1.py"
Current URL: https://www.saucedemo.com/inventory.html
PASS: Login successful
PASS: URL contains /inventory.html
PS C:\Users\catch\OneDrive\Desktop\wipro_assignments>
```

7.Result :-

The Selenium login test was successfully designed to automate the SauceDemo login process using multiple locator strategies and to verify the expected post-login URL.

8.Observation :-

The username, password, and login button were identified using ID, NAME, and XPath locators respectively. The URL assertion can be used to confirm successful navigation after login.

9.Conclusion :-

Thus, Selenium WebDriver was successfully used to automate a login workflow using different locator strategies and an assertion to validate the resulting page URL.